Adaptive Question Selection Using Upper Confidence Bound: A Simulation Study

1. Introduction and Objectives

This project investigates whether **bandit-based adaptive question selection** can improve student learning outcomes compared to uniform random selection in an educational quizzing context. Specifically, we apply the **Upper Confidence Bound (UCB1)** algorithm — originally developed for the multi-armed bandit problem — to dynamically choose which knowledge category to quiz a student on next.

The central research question is:

> *Does UCB-guided adaptive question selection produce faster, more balanced knowledge acquisition than random selection, particularly for a student's weakest knowledge areas?*

The hypothesis is that by treating each knowledge category as an "arm" and using the UCB exploration-exploitation trade-off, the system can prioritize weak areas while still maintaining coverage of all topics — leading to more efficient learning.

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2. Method

2.1 Student Model

Each simulated student maintains a **knowledge vector** $\mathbf{K} = [k_1, k_2, \ldots, k_6]$, where $k_i \in [0, 1]$ represents the probability of answering correctly in category i. Initial knowledge is sampled from $k_i \sim \mathcal{N}(\mu \cdot d_i,; \sigma^2)$, where d_i is the category difficulty and $\mu = 0.35$, $\sigma = 0.1$.

Learning dynamics. When a student answers a question in category i:

- **Correct answer:** $k_i \leftarrow k_i + \alpha \cdot (1 - k_i)$, an asymptotic update toward 1.0 (learning rate $\alpha = 0.12$).
- **Incorrect answer:** $k_i \leftarrow k_i - \beta \cdot k_i$, a multiplicative penalty ($\beta = 0.02$).

Forgetting dynamics. At each timestep, unexposed categories decay following an Ebbinghaus-inspired model:

$$k_i \leftarrow k_{\text{base}} + (k_i - k_{\text{base}}) \cdot e^{-\lambda}$$

where $k_{\text{base}} = 0.10$ is the retention floor and $\lambda = 0.01$ is the decay rate.

2.2 Question Selection Strategies

Two selection strategies are compared in a between-subjects simulation design:

UCB Group (Adaptive). The next category to quiz is selected by maximizing:

$$\text{UCB}(i) = \underbrace{(1 - r_i)}{|text{exploitation}} + |underbrace{c |cdot |sqrt{|frac{|ln N}{n_i}}} {\text{exploration}}$$

where r_i is the correctness rate for category i, N is the total number of questions asked, n_i is the number of attempts on category i, and $c = \sqrt{2} \approx 1.414$ is the exploration parameter. The

first term prioritizes categories with low correctness (weak areas); the second term encourages exploration of under-tested categories.

Control Group (Random). Categories are selected uniformly at random with probability $1/6$ each.

2.3 Experiment Design

Parameter	Value
Students per group	100
Knowledge categories	6
Questions per student	10,000
Learning rate (α)	0.12
Incorrect penalty (β)	0.02
Forgetting decay rate (λ)	0.01
Base knowledge floor (k_{base})	0.10
UCB exploration constant (c)	1.414
Random seed	42 (for reproducibility)

Both groups share identical student populations (same initial knowledge distributions and random seeds) to isolate the effect of the selection strategy. The experiment was replicated across 3 independent runs. Group-level metrics (mean knowledge, weakest category performance, knowledge variance, cumulative accuracy) were recorded at every timestep.

2.4 Analysis Framework

Results are analyzed across three temporal phases:

- **Short-term** (0–500 questions): Initial learning and exploration.
- **Mid-term** (500–2,000 questions): Knowledge consolidation.
- **Long-term** (2,000–10,000 questions): Saturation and convergence behavior.

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3. Results

3.1 Overall Knowledge Acquisition

UCB achieves higher average knowledge during the short-to-mid term but random selection converges to comparable (and slightly higher) levels given sufficient time.

Timestep	UCB Avg Knowledge	Random Avg Knowledge	UCB Advantage
100	0.321	0.348	-7.6%
500	0.566	0.544	**+3.9%**
1,000	0.588	0.570	**+3.1%**
2,000	0.569	0.563	**+1.1%**

5,000	0.547	0.568	-3.7%
10,000	0.536	0.562	-4.6%

UCB's peak overall advantage of **+2.3%** occurs at $t \approx 470$. UCB leads in overall knowledge for **21.2%** of all timesteps, concentrated in the first 2,000 questions.

3.2 Weakest Category Protection

The most striking difference is in **weakest category performance**, where UCB shows a dramatic early advantage:

Timestep	UCB Weakest	Random Weakest	UCB Advantage
100	0.207	0.129	**+60.7%**
250	0.410	0.262	**+56.7%**
500	0.500	0.365	**+36.8%**
1,000	0.509	0.407	**+25.2%**
2,000	0.458	0.398	**+15.0%**
5,000	0.390	0.386	+0.8%
10,000	0.358	0.390	-8.2%

UCB's peak weakest-category advantage is **+60.7% at $t = 100$**, and it leads for **53.5%** of all timesteps. This confirms that UCB is highly effective at rapidly lifting the lowest-performing knowledge area.

3.3 Phase Summary

Phase	UCB Overall Advantage	UCB Weakest Category Advantage	Winner
Short-term ($t = 500$)	**+3.9%**	**+36.8%**	**UCB**
Mid-term ($t = 2{,}000$)	**+1.1%**	**+15.0%**	**UCB**
Long-term ($t = 10{,}000$)	-4.6%	-8.2%	Random

3.4 Category-Level Observations

- **UCB** produces more uniform knowledge heatmaps across categories, indicating balanced learning.
- **Random** shows higher inter-category variance early on, with some categories advancing much faster than others.
- Final category knowledge range: UCB = 0.411, Random = 0.320. UCB has wider spread at termination due to its aggressive earlier focus on weak categories creating temporary imbalances during the long-term phase.

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4. Discussion and Key Observations

1. **UCB excels in time-constrained settings.** The adaptive strategy provides 25–60% better protection for weak areas during the first 500–1,000 questions. For practical learning scenarios where study time is limited, this is a significant advantage.

2. **Convergence under extended practice.** Given enough questions (~10,000 across 6 categories, or ~1,667 per category), random selection achieves comparable and slightly higher overall knowledge. This is expected: with sufficient exposure, even uniform allocation provides adequate coverage.

3. **The exploration-exploitation trade-off works as intended.** UCB's initial focus on weak categories comes at a small short-term cost to overall accuracy (UCB is -7.6% at $t = 100$) but pays off by $t = 500$ when it surpasses random in overall knowledge.

4. **UCB's value lies in efficiency, not ceiling.** Both strategies reach similar knowledge ceilings (~0.55–0.59). UCB reaches this level faster and with more balanced category coverage.

5. **Diminishing returns of adaptivity.** As all categories receive sufficient practice, UCB's advantage diminishes because the exploitation signal (correctness rate differences) shrinks, and the exploration bonus becomes negligible relative to accumulated experience.

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5. Preliminary Conclusions

- The UCB algorithm is an effective and principled approach for adaptive question selection, particularly when learning time is constrained.
- UCB's primary benefit is **learning efficiency** — achieving balanced, higher knowledge in fewer questions — rather than improving the absolute knowledge ceiling.
- For sessions under ~1,000 questions per student, UCB is strongly recommended over random selection.
- Future work could explore: (a) hybrid strategies (UCB early, random later), (b) tuning the exploration parameter c for different learner profiles, (c) non-stationary variants that account for changing student knowledge, and (d) validation with real student data.

Experiment conducted with random seed 42 for reproducibility. Full source code, data, and visualizations are available in the project repository.